1. **MS:** 5.4-5, 16.1, 16.2

2. **Additional Questions:**

   (a) A consumer has the utility function $U(x, y) = xy$, where x are shirts and y are hats. This consumer faces a budget constraint of $100. The price of $x = 2$ and the price of $y = 1$. Find the number of hats and shirts that maximizes this consumer's utility.

   (b) Benny consumes coffee (c) and bagels (b). He has an income of 42. Bagels are sold in an unusual way. There is only one supplier, and the more bagels Benny buys, the higher the price per unit. In fact, bagels cost Benny a total of $2b^2$ for b bagels.[1] Coffee is sold normally at a price of $2 per unit. Benny's utility function is $u(b, c) = 12b + 4c$. Depict Benny's budget line and a few generic indifference curves on a graph and then find the optimal bundle of coffee and bagels that maximizes Benny's utility.

[1] If you buy three bagels, the total expenditure on the bagels is $2(3)^2$=$18.